

Tim van Beeck

Ph.D. Student · Numerical Mathematics

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Scientific biography

I am a Ph.D. student at the Institute for Numerical and Applied Mathematics, University of Göttingen, supervised by Christoph Lehrenfeld. My research centers on the design and numerical analysis of finite element methods for partial differential equations. My current focus is on the modeling, analysis, and numerical simulation of time-harmonic wave propagation in nematic liquid crystals (nematoacoustics). Previously, I worked on the numerical analysis of Galbrun's equation arising in helioseismology. I have also contributed to the development of structure-preserving discretizations for unfitted mixed finite element methods, the axisymmetric Stokes problem, and Navier–Stokes flows on surfaces.

Education

since 04/2024	**Ph.D. in Mathematics**, University of Göttingen, Germany. *Supervisor:* Christoph Lehrenfeld.
12/2021 – 12/2023	**M.Sc. in Mathematics**, University of Göttingen, Germany, Grade: 1.0 (with distinction). *Thesis: On stable discontinuous Galerkin discretizations for Galbrun's equation.* *Supervisor:* Christoph Lehrenfeld.
10/2017 – 12/2021	**B.Sc. in Mathematics**, University of Göttingen, Germany.
2009 – 2017	**Secondary school** (Abitur), Gymnasium Corvinianum Northeim, Germany.

Employment history

Positions:

since 01/2024	**Research associate**, University of Göttingen, Germany.
11/2021 – 12/2023	**Student assistant**, University of Göttingen, Germany.

Teaching:

04/2026 – 09/2026	Mathematically oriented programming (University of Göttingen, lecture and tutorial class, 4 h per week).
10/2025 – 03/2026	Mathematically oriented programming (University of Göttingen, tutorial class, 2 h per week).
10/2022 – 03/2023	Partial differential equations (University of Göttingen, tutorial class, 2 h per week).

Grants, awards, and honors

2027 – 2029	Elected member of the GAMM Juniors.
10/2026	DAAD "Forschungsstipendien für Doktorandinnen und Doktoranden" (grant to finance a research stay at the University of Oxford, ∼3,000 EUR).
05/2025 – 05/2026	GAMM Junior Mentoring Program (mentored by Benedikt Gräßle).

Publications

Journal articles:

[4] T. Brüers, C. Lehrenfeld, **T. van Beeck**, and M. Wardetzky (2026). *Releasing the pressure: High-order surface flow discretizations via discrete Helmholtz–Hodge decompositions.* In: Computer Methods in Applied Mechanics and Engineering, Vol. 461, DOI: 10.1016/j.cma.2026.119201.

[3] E. Burman, J. Preuss, and **T. van Beeck** (2026). *Variational data assimilation for the wave equation in heterogeneous media.* In: Communications on Applied Mathematics and Computation, DOI: 10.1007/s42967-026-00583-w.

[2] M. Halla, C. Lehrenfeld, and **T. van Beeck** (2026). *Hybrid discontinuous Galerkin discretizations for the damped time-harmonic Galbrun's equation.* In: ESAIM: Mathematical Modelling and Numerical Analysis, Vol. 60, 1135–1175, DOI: 10.1051/m2an/2026027.

[1] C. Lehrenfeld, **T. van Beeck**, and I. Voulis (2025). *Analysis of divergence-preserving unfitted finite element methods for the mixed Poisson problem.* In: Mathematics of Computation, Vol. 94, 1667–1699, DOI: 10.1090/mcom/4027.

Preprints:

[4] **T. van Beeck** (2026). *A hybrid C^0-interior penalty method for the nematic Helmholtz–Korteweg equation.* arXiv:2606.25642.

[3] P. L. Lederer, C. Lehrenfeld, C. Merdon, and **T. van Beeck** (2026). *Pressure-robustness for the axisymmetric Stokes problem by velocity reconstruction.* arXiv:2603.01030.

[2] P. E. Farrell, **T. van Beeck**, and U. Zerbinati (2025). *Analysis and numerical analysis of the nematic Helmholtz–Korteweg equation.* arXiv:2503.10771.

[1] **T. van Beeck** and U. Zerbinati (2024). *An adaptive mesh refinement strategy to ensure quasi-optimality of finite element methods for self-adjoint Helmholtz problems.* arXiv:2403.06266.

Scientific exchange

Research visits (one week, unless stated otherwise):

10/2026 – 12/2026	Research stay at the University of Oxford, United Kingdom, working with P. E. Farrell.
08/2026	Simula Research Laboratory, Norway, working with U. Zerbinati.
04/2026	University of Vienna, Austria, working with P. Stocker.
10/2025 & 03/2026	University of Oxford, United Kingdom, working with U. Zerbinati & P. E. Farrell.
01/2024 – 03/2024	Research stay at University College London, United Kingdom, working with E. Burman & J. Preuss.

Conference talks (selected out of 14):

09/2026	Advances in Ordered Fluids and Alignment Phenomena, Golfrenzo, Italy. *Tuning sound with liquid crystals: FEM for nematoacoustics.*
07/2026	CMAM 2026, Vienna, Austria. *A hybrid C^0-interior penalty method for the nematic Helmholtz–Korteweg equation.*
09/2025	ENUMATH 2025, Heidelberg, Germany. *Divergence-preserving unfitted FEM for the mixed Poisson problem.*
07/2024	WCCM 2024, Vancouver, Canada. *Divergence-preserving unfitted FEM for the mixed Poisson problem.*
06/2024	WAVES 2024, Berlin, Germany. *Stable hybrid discontinuous Galerkin discretization for Galbrun's equation.*

Administrative work

04/2024 – 03/2027	Member of the managing board of the Institute for Numerical and Applied Mathematics (representing scientific employees).
since 10/2025	Member of the selection committee for the "Deutschlandstipendium".
04/2025 – 03/2026	Deputy Ph.D. students' representative (Faculty of Mathematics & Computer Science).